Experiment -2

Python Program To Solve 8-Queen Problem

Code:

```
N = 8
board = [-1] * N

def safe(row, col):
    for i in range(row):
        if board[i] == col or abs(board[i] - col) == abs(i - row):
            return False
    return True

def solve(row):
    if row == N:
        for i in range(N):
            for j in range(N):
                print("Q" if board[i] == j else ".", end=" ")
            print()
        return True

    for col in range(N):
        if safe(row, col):
            board[row] = col
            if solve(row + 1):
                return True
            board[row] = -1
```

return False

if not solve(0):

print("No solution")

```
N = 8
board = [-1] * N

def safe(row, col):
    for i in range(row):
        if board[i] == col or abs(board[i] - col) == abs(i - row):
            return False
    return True

def solve(row):
    if row == N:
        for i in range(N):
            for j in range(N):
                print("Q" if board[i] == j else ".", end=" ")
            print()
        return True

    for col in range(N):
        if safe(row, col):
            board[row] = col
            if solve(row + 1):
                return True
            board[row] = -1

    return False

if not solve(0):
    print("No solution")
```

```
Q . . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .


...Program finished with exit code 0
Press ENTER to exit console.
```